YUTIKA AGARWAL

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SUMMARY

Artificial Intelligence / **Machine Learning** professional with a strong foundation in **Machine Learning**, **Deep Learning**, and **Advanced Data Analytics**, with hands-on experience in **NLP**, **Computer Vision**, and **Neural Networks**. Proficient in **Python**, **TensorFlow**, **PyTorch**, **Keras**, **SQL**, and **data visualization** tools, and passionate about applying **AI** to solve real-world problems.

EDUCATION

Master of Engineering – Artificial Intelligence (GPA: 3.96/4.00) Aug 2024 – Apr 2026
University of Cincinnati – Cincinnati, Ohio, USA

Bachelor of Technology – Computer Science & Engineering (CGPA: 8.28/10.00) Jul 2020 – May 2024
The NorthCap University – Gurugram, Haryana, India

SKILLS

Programming & Databases: Python, SQL
AI & Machine Learning: Deep Learning, Neural Networks, Machine Learning, NLP, Predictive Modeling, Object Detection, OCR, Image Segmentation, LLM Fine-Tuning, RAG, Agents
Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, Keras, Pandas, NumPy, OpenCV, Ultralytics (YOLO), NetworkX, Hugging Face Transformers, LangChain, FAISS, Qdrant, PyMuPDF, Ollama
Data Visualization: Power BI, Matplotlib, Seaborn
Tools & Platforms: Roboflow, AutoCAD, Git, GitHub, Docker, Streamlit

WORK EXPERIENCE

Neilsoft Pune, India
AI/ML Engineer May 2026 – Present

- Developing an AI-driven plan-check automation pipeline that ingests architectural PDF plan sets and answers building-compliance checklist questions (**177**-rule checklist) for AEC engineering workflows.
- Built a PDF triage classifier categorizing drawing pages as native-text, **OCR**-layer, scanned, or outlined; implemented an **OCR pipeline** (**RapidOCR**/**Tesseract**) achieving near-complete sheet-number extraction across a **17**-document plan-set corpus, plus vector-path extraction using **PyMuPDF**.
- Deployed and orchestrated local **LLMs** (**Llama**, **Gemma** via **Ollama**) and built a **Python** agent to route queries between models for on-premises AI workflows.

OptifloAI Solutions Abu Dhabi, UAE
AI/ML Engineer Jan 2026 – Apr 2026

- Built and fine-tuned **YOLOv11l**/**YOLOv8l object detection** models for automated **P&ID** symbol recognition (**~88%** mAP50), engineering end-to-end ML pipelines including **synthetic data** generation from **DXF**/**AutoCAD**, image tiling, **Roboflow** annotation, **OCR** text extraction, graph-based connectivity modeling with **NetworkX**, and GPU-accelerated training on Google Colab.
- Built a custom code-generation model by fine-tuning **Llama 3.1 8B** with **QLoRA**, cutting memory footprint by **75%** (**16GB → 4GB**) via **4-bit quantization**, and training a **20MB LoRA** adapter on **18.6K** samples that produces syntactically correct **Python** for algorithm implementation, file I/O, and data processing.

CDAC (Centre for Development of Advanced Computing) Bangalore, Karnataka, India
AI Researcher Intern Jan 2024 – May 2024

- Developed and evaluated ML models for "**Uncertainty Quantification**" using **Python**, **TensorFlow**, and **PyTorch** to enhance model reliability and robustness under adversarial conditions.
- Applied uncertainty estimation, data preprocessing, feature engineering, and performance analysis to improve the trustworthiness and interpretability of AI systems.

Doon University Dehradun, Uttarakhand, India
Research Intern Jun 2022 – Jul 2022

- Contributed to "Designing Hybrid Introspection Enabled **VMM**-based Security Architecture" by building and configuring virtual environments, conducting experimental studies, and integrating **virtualization** with security concepts to support architecture design and validation.

PROJECTS

- **PaperSage – AI Assistant for Research Papers (Live Demo · GitHub):** Built and deployed a live web app that lets anyone ask plain-English questions about **200** machine-learning research papers and get clear, source-cited answers with no made-up facts (**Retrieval-Augmented Generation**, **RAG**). Combined keyword and AI-based search with a reranking model I fine-tuned to improve accuracy (top result **76% → 83%**); also understands uploaded images and has user login.
- **VaultIQ – Credit Risk Analytics on 2.2M Loans (GitHub):** Analyzed **2.26 million** real loans from start to finish — cleaned the messy raw data with **Python**, stored it in a **MySQL** database, used **SQL** to study why loans go unpaid, and built an easy-to-read **Power BI** dashboard. Found that high-risk grades and small-business loans fail far more often (from **~6%** up to **~50%**).
- **Medical Image Segmentation (Retinal Blood Vessel Detection) (GitHub):** Implemented 2- and 3-layer **U-Net** architectures on **100** retinal images (**80** train / **20** test) with preprocessing, augmentation, and optimization (Adam, LR scheduling), achieving best results with the 2-layer **U-Net** (Dice **0.74**, IoU **0.58**).
- **Boruta-based Feature Selection for Heart Disease Prediction (GitHub):** Built a **Boruta**-based **feature selection** model; **Random Forest** achieved **88%** accuracy and **0.8841 ROC AUC**, with comparative analysis against **Lasso**, **Ridge**, and other algorithms.

PUBLICATIONS

- Agarwal, Y., Chhikara, R., Rana, S. (2023). **Boruta** based **feature selection** model for heart disease prediction.
- Nautiyal, S., Saklani, A., Pant, A., Agarwal, Y., Gaur, A., Mishra, P. "VNSecure: An explainable virtual network attack detection framework at **VMM**-Layer," 2023 Fifteenth Intl. Conf. on Contemporary Computing, pp. 1–**12**.
- Bhatt, M., Gaur, A., Badoni, S., Agarwal, Y., Mishra, P. "Advanced malware and their impact on **virtualization**: hybrid feature extraction using deep **memory introspection**," 2022 Fourteenth Intl. Conf. on Contemporary Computing, pp. **74**–**80**.

CERTIFICATIONS

AI Agent and Agentic AI Software Development (Coursera) • Introduction to Generative AI (Google) • Explore Machine Learning Models with Explainable AI • AWS Academy Graduate – Cloud Foundations